

Riley Brown

rydb@proton.me · rbrown71@nebraska.edu · ry3b.github.io · github.com/ry3b · 531-333-1615

Education

University of Nebraska-Lincoln

B.S. Mathematics and Data Science, Minor in Computer Science

- GPA 3.92/4.00
- University Honors Program

Expected May 2023–May
2027
Lincoln, NE

Budapest Semesters in Mathematics

Study abroad, mathematics

Jan 2026–Jun 2026
Budapest, Hungary

Research Experience

Undergraduate Researcher - Quantum LDPC Codes

University of Nebraska-Lincoln advisor: *Professor Christine Kelley*

Fall 2026
Lincoln, NE

Research Assistant — pseudorandomness

North Carolina State University advisor: *Professor Chin Ho Lee*

Summer 2026
Raleigh, NC

- Worked through Ta-Shma's construction of epsilon-balanced codes and the surrounding literature as a research partner with Prof. Chin Ho Lee rather than in a supervised role
- Developed simpler constructions of epsilon-balanced codes
- Made minor improvements to the construction of Hsieh, Mohanty and Zheng

Research Assistant — 3-uniform hypergraphic degree sequences, FPT algorithm feasibility

Budapest Semesters in Mathematics advisor: *Professor Istvan Miklos*

Spring 2026
Budapest, Hungary

Summer Researcher — algorithm design

Harvey Mudd College advisor: *Professor Tim Randolph*

Summer 2025
Claremont, CA

- Collaborated with a team of four students to analyze and improve exponential-time algorithms for the sumset recognition problem, an NP-complete problem in additive combinatorics
- Enhanced branching approaches and refined base case parameters for graph-theoretic reduction methods through systematic analysis of algorithmic complexity
- Presented findings at the REU symposium and continued independent exploration of the problem space

Investigating the Flow of Opinions in a Diagonal Grid Graph Using a Modified Deffuant-Weisbuch Model

University of Nebraska-Lincoln, Directed Reading Program advisor: *Nick White*

Fall 2024
Lincoln, NE

- Extended the project beyond its initial reading goals by implementing a modified Deffuant-Weisbuch model in Python
- Ran systematic parameter variation experiments showing that opinions invariably converge to consensus on diagonal grid graphs
- Proved the discovered properties and identified future directions on alternative graph topologies

Undergraduate Researcher — tiling theory

University of Nebraska-Lincoln advisor: *Professor Tri Lai*

Summer 2024

Lincoln, NE

- Applied Ciucu’s Factorization Theorem to decompose tiling enumeration problems into tractable subproblems, working up through independent study of survey papers
- Used Randall paths to determine hexagon-half parameters based on hole parity along the symmetric axis
- Computed exact tiling counts by applying the Lindstrom-Gessel-Viennot theorem to determinants of binomial coefficient matrices

Preprints and Publications

Fixed-Parameter Tractability of t-Uniform Hypergraphicality

Riley Brown, Istvan Miklos

arXiv preprint, 2026

arXiv

Awards and Honors

D&F Eastman Scholarship

Mathematics Department recognition

Regents Scholarship

Nebraska Career Scholarship

Dean’s List

all semesters

University Honors Program

Talks

Algorithms for the Sumset Recognition Problem

2025

Harvey Mudd College REU Symposium

Combinatorial Game Theory

2025

UNL Directed Reading Program Symposium

Mathematics at UNL

2025

Marshall Recruitment Talk for High School Students

The Modified Deffuant-Weisbuch Model

2024

UNL Directed Reading Program Symposium

Lozenge Tilings of a Hexagon with Holes

2024

UNL Math Club

Teaching

Tap Dance Instructor

2024

Chase the Spirit of Dance, Lincoln NE and Create 308, Grand Island NE (2018-2024)

Teaching Assistant, Calculus II

Learning Assistant, Analysis

Teaching Assistant, Honors Calculus II

Projects

Poker Trainer

python · 2026

A terminal and Tkinter drill trainer for poker skills: pot odds, combos, equity, bet sizing, push/fold and ICM, with an adaptive scheduler that weights drills by weakness and staleness.

Selected Coursework

Mathematics. Bijective Partition Theory, Discrete Mathematics, Matroid Theory, Advanced Combinatorics, Conjecture and Proof, Discrete and Convex Geometry, Graph Theory, Group Theory, Probability Theory, Stochastic Processes, Elementary Analysis, Intro to Modern Algebra, Linear Algebra, Differential Equations, Multivariable Calculus, Math in the City, **Algebra I**, **Coding Theory**

Computer Science. Development and Analysis of Algorithms, Automata, Computation, and Language, Data Structures and Algorithms, Data Analysis, Math for Machine Learning, Generative AI, Computational Game Theory

Skills

Languages. Python, MATLAB, Java, SQL, LaTeX

Libraries and tools. NumPy, SciPy, Pandas, Matplotlib, scikit-learn, PyTorch, TensorFlow, Git, Jupyter